MMF 1921 Final Exam May 12, 2023

INSTRUCTIONS: *In general, SHOW ALL WORK. Exam is CLOSED book and CLOSED notes except for both sides of half an 8.5 inch by 11 inch sheet of handwritten notes. A non-programmable scientific calculator is allowed. There will be no questions allowed during the exam. You must turn in the exam questions with your answer booklet. Note: for all problems that require you to formulate a model you must define all variables and parameters.*

Problem 1 (15 points)

You have the following three investments available to you to invest money in.

Investment A: For each dollar invested in A at time 0, we receive \$0.10 at time 1 and \$1.30 at time 2. (Time 0 = now; time 1 = one year from now, and so on.)

Investment B: For each dollar invested in B at time 1, we receive \$1.60 at time 2.

Investment C: For each dollar invested in C at time 2, we receive \$1.20 at time 3.

At any time, leftover cash may be invested in T-bills, which pay 10% per year. At time 0, we have \$100. At most, \$50 can be invested in each of investments A,B, and C. Formulate (but NOT solve) a linear program that can be used to maximize your cash at time 3.

Problem 2 (20 points)

Consider the LP

$$\begin{array}{ll} \text{minimize} & -4x_1 - 3x_2 - 2x_3 \\ \text{subject to} & 2x_1 + 3x_2 + 2x_3 \leq 6 \\ & -x_1 + x_2 + x_3 \leq 5 \\ & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0 \end{array}$$

(a) (5 points) Write the dual of this LP.

(b) (15 points) Solve the primal problem using the simplex method and show that at optimality the dual problem is solved to optimality as well.

Problem 3 (20 points)

Part 1

(6 points) Consider the function

$$f(x) = 2x_2^3 - 6x_2^2 + 3x_1^2 x_2$$

Find all stationary points and classify them according to whether they are saddle points, strict/non-strict local/global minimum/maximum points.

Part 2

Consider the function $f(x) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$.

(a) (2 points) Prove that $x = \binom{1}{1}$ is the unique global minimizer of $f(x)$ over $\mathbb{R}^2$

(b) (6 points) With an initial point $x^{(0)} = \binom{0}{0}$ apply two iterations of Newton's method for minimizing $f(x)$ over $\mathbb{R}^2$.

(c) (6 points) With an initial point $x^{(0)} = \binom{0}{0}$ apply two iterations of gradient method for minimizing $f(x)$ over $\mathbb{R}^2$ using a constant step length of 0.05.

Problem 4 (10 points)

Consider the minimum variance problem (i.e. the problem of minimizing portfolio variance subject to only budget constraints with short selling allowed) over n risky assets. Let V be the covariance matrix of the n assets. The model can be written as min $x^T V x$ subject to $e^T x = 1$ where e is a vector of all ones.

(a) (5 points) Find a closed-form solution for the optimal minimum variance portfolio. State the conditions under which the closed-form solution is well defined and unique and justify your answer.

(b) (5 points) Now consider the minimum variance problem but with no short selling allowed. Find the KKT conditions for this problem. Is a closed-form solution possible in this case? If so find a closed-form solution or explain why it is not possible.

Problem 5 (12 points, 3 points each, To get full credit for a part you must have the correct True or False answer and valid justification)

Note: Give a **brief** justification for your answer.

(a) True or False: Risk Parity portfolios do not always have better risk-return characteristics compared to minimum variance portfolios.

(b) True or False: For a non-linear optimization problem with both inequality and equality constraints, at optimality the optimization problem can be essentially transformed to a problem with only equality constraints.

(c) True of False: In the Black-Litterman framework presented in class, subjective views are allowed to be correlated.

(d) True or False: The KKT conditions are both sufficient and necessary for all convex optimization problems.

Problem 6 (13 points)

You are analyzing an investment opportunity with two stocks A and B with the expected return of Stock A $\mu_A = 0.025$, the expected return of Stock B $\mu_B = 0.015$, the variance of Stock A $\sigma_A^2 = 0.0049$, the variance of Stock B $\sigma_B^2 = 0.0025$, and $\sigma_{AB} = -0.001$.

Suppose that the expected returns were estimated using the last 20 monthly observations. Now consider the following robust optimization problem with an ellipsoidal uncertainty set

$$
\begin{array}{ll}
\min\limits_{x,y} & x^T Q x \\
\text{s.t.} & \mu^T x - y \geq 1\% \\
 & y^2 \geq x^T \Theta x \\
 & 1^T x = 1 \\
 & y \geq 0
\end{array}
$$

Use this information to check whether the solution $x = [0.425\ 0.575]^T$ satisfies the KKT conditions of the problem above. (Note: Assume the target return constraint is active.)

Problem 7 (10 points)

Below is the scenario-based CVaR minimization formulation

$$
\begin{array}{lll}
\text{minimize} & \gamma + \frac{1}{(1-\beta)S} \sum\limits_{s=1}^{S} z_s & \\
 & z_s \geq 0 & s = 1, , , S \\
 & z_s \geq f(x, y_s) - \gamma & s = 1, , , S \\
 & 1^T x = 1 & \\
 & \mu^T x \geq R & \\
 & x \geq 0 &
\end{array}
$$

(a) (5 points) Define γ, β, z_s, y_s, and $f(x, y_s)$.

(b) (5 points) Formulate the most tractable robust counterpart of the model above where the model should be robust with respect to μ by using ellipsoidal uncertainty sets. Define all quantities you need for the robust formulation that are not already in the formulation above.